
ENGINEERING SPECIFICATIONS

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Chiller Plant Manager

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1. Introduction

Aspiration for energy conservation and necessity to reduce manual intervention in plant operation to minimum level, demands for Plant Manager (PM). At the same time the recent developments in programmable controllers and control equipment resulting in low cost and high reliability made it practically feasible.

Kirloskar Chillers Pvt. Ltd. has developed the Plant Manager, flexible enough to accommodate various combinations of plant equipment & there operating sequences and smart enough to stage/operate them to optimize for power and performance.

2. Functionality

Types of CPM designs

- 1) Primary Constant System
- 2) Primary Variable System
- 3) Primary Secondary System

A. General

Chillers will be operated in local mode & remote mode; local/remote mode selection shall be done from chiller controller. Remote mode operation shall be initiated from Plant-Manager (PM) by start/stop command to chiller control panel. During Local Mode Plant-Manager (PM) intervention shall be bypassed.

Chillers will be sorted based on run-hours and healthy feedback of them along with all allied items (pumps, valves etc.). Trip feedback of any one item for corresponding chiller, makes it unavailable. First number will be assigned to chiller with least run-hours. Start- up of chiller in algorithm will also mean starting corresponding pumps & cooling tower.

Chilled water isolation valve & Chiller are together considered as one group since failure of any one will make other unavailable, whereas two such groups are independent of each other for healthiness. Same is the case with evaporator pumps since they are connected with each other via common header system, so there can be any pump associated with any group of chiller & valve. Alarm condition (Trip, fail to Start/Open and Fail to Stop/Close) of any element in group is considered as alarm for that group and under such conditions that particular group is immediately replaced by next available alternate group.

B. Staging of Chillers

Single set point and a dead band are set to decide loading and unloading profile with respect to supply water temperature value. Dead band is equally distributed across setpoint so that loading zone is for supply water temperature $> \text{Setpoint} + \text{Dead band}/2$ while unloading zone is for supply water temperature $< \text{Setpoint} - \text{Dead band}/2$. While temperature is in dead band, no loading or unloading is done.

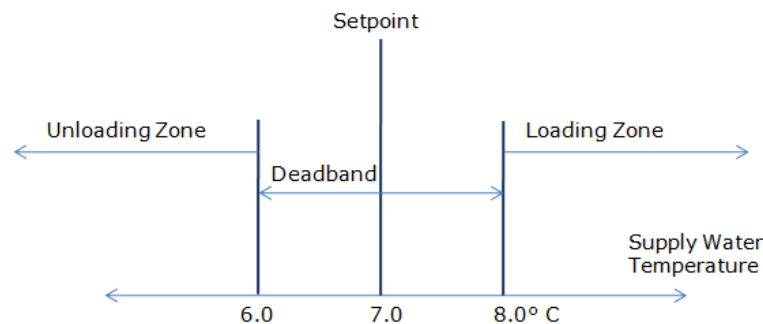


Fig.1 Chiller Setpoint

1. Loading sequence for Chillers

Plant manager needs start/ Auto command to start the operation. Once it is started, status of all the equipment is evaluated and correct sequence is set. In case of chillers, sequence is set in such a way that the chiller with lowest run-hours is set at first to start and others in ascending order ahead. After starting up, one evaporator pump along with evaporator valves of first chillers are turned on irrespective of demand. If temperature is in loading zone after startup, demand is generated for chillers one by one using PID control logic. Demand for next chiller is generated only when already running chillers cross set limit of loading or run for preset time and delay time between demand steps is elapsed. Staging of pumps will directly be governed by staging of chillers.

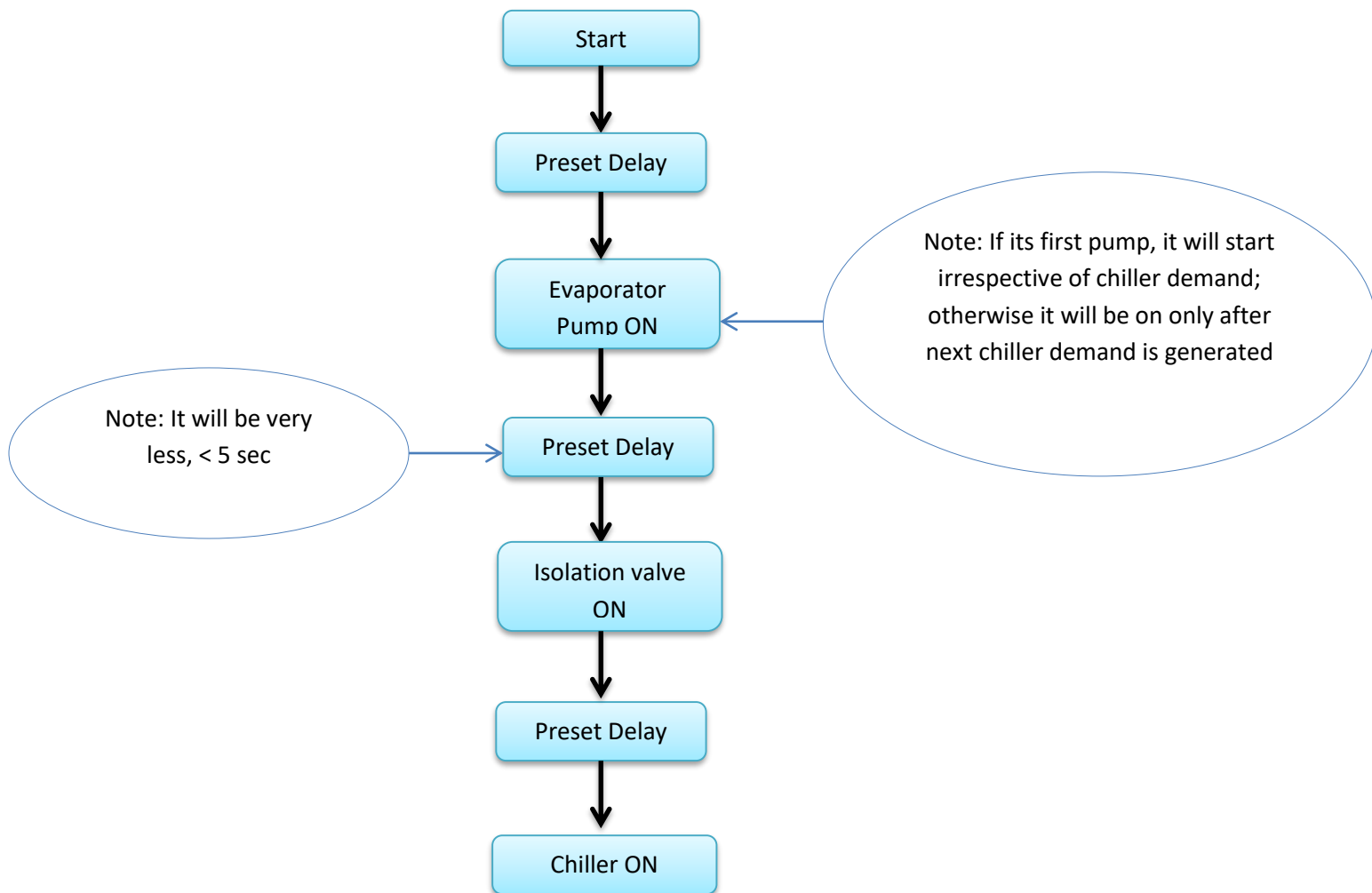


Fig.2 Loading Sequence

2. Unloading sequence for chillers

When supply water temperature is in unloading zone chiller units are turned off one by one in such a way that the chiller with highest run-hours is turned off first and others in descending order ahead. If temperature is in Unloading zone, demand is removed for chillers one by one using PID control logic. Selected chiller is turned off only when running chillers are unloaded to preset level. After turning off one unit there is delay time which must be elapsed before checking condition for turning off next chiller. Even if chiller is OFF corresponding valve and pump will be ON for preset delay.

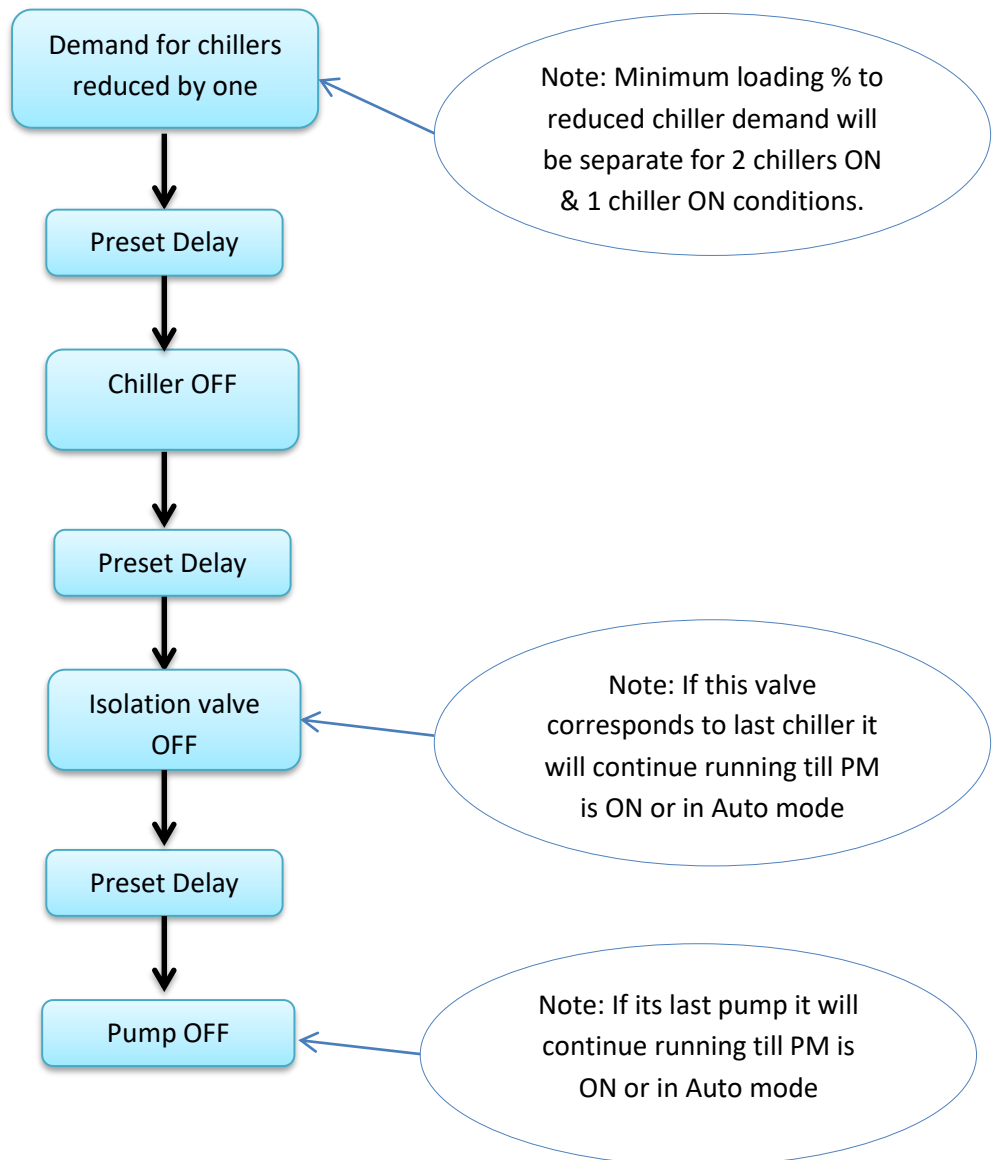


Fig.3 Unloading Sequence

C. Modulation of Pumps

Primary variable system is used to operate plant at various load, hence all evaporator pumps have VFD controlled by individual pump PLC. Pump PLC will be interfaced with CPM and takes command for staging of pumps as well as communicate back the status of pumps to CPM.

The plant contains variable primary water loop circulating through chiller, primary pumps and load. As common headers are used for chilled water, motorized isolation valves are provided at each chiller outlet. There are 3 primary pumps used for 2 evaporators. VFDs are used to vary primary pump speeds which are operated based on feedback of actual load from AHU. For sequence operation the communication to be established between pump PLC and Plant Manager. It is advisable that the communication with pumps and Plant Manager should be hardwired for safe operation, but soft communication is also possible if hardwired is not available.

D. Bypass valve Operation

One essential thing in this design is providing bypass line, from supply header to return header, which contains automated modulating valve. CPM will continuously monitor chilled water supply header flow by flow meter to ensure minimum flow required to keep chillers running. CPM will calculate minimum flow required based on number of chillers ON and if pump PLC reduces flow up to minimum requirement then CPM will start modulating bypass valve.

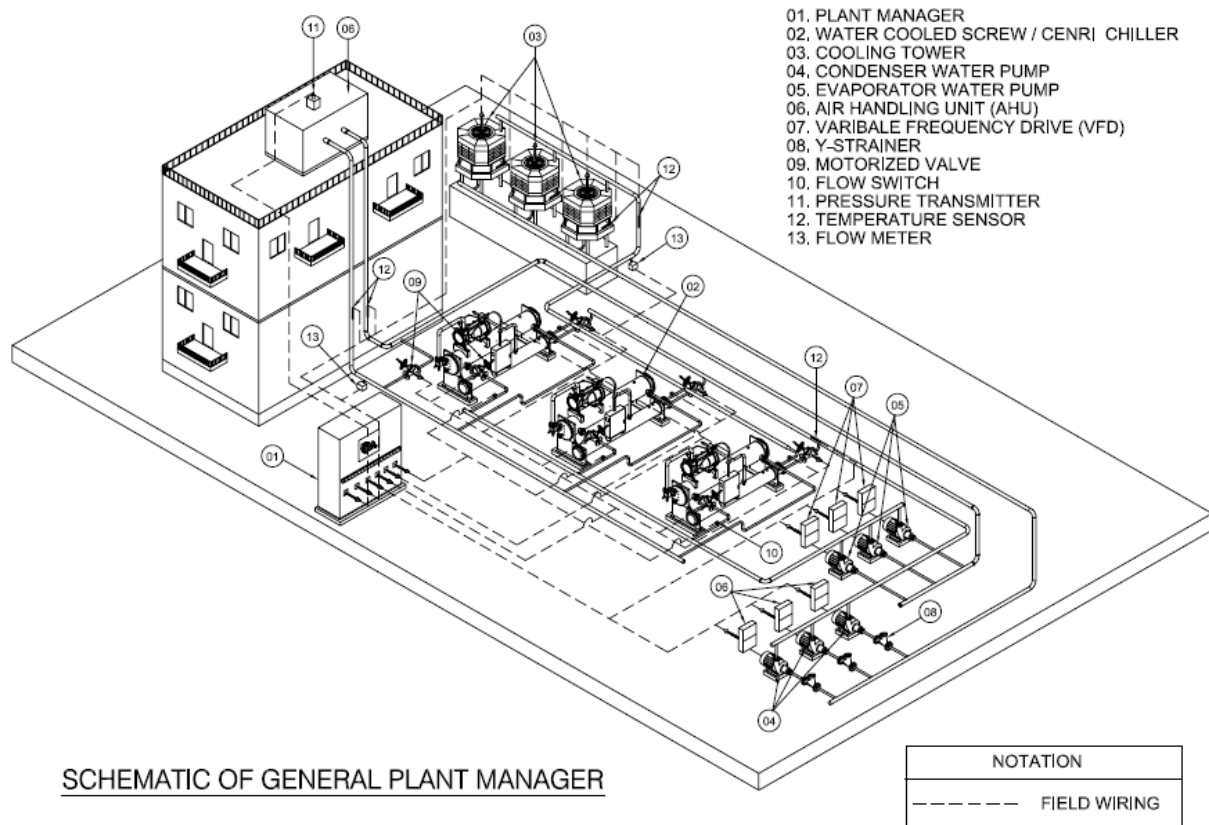
3. Plant Description

Chiller Plant Manager (CPM) controls/operates following equipment in plant,

- a. Air/Water cooled chillers
- b. Evaporator pump for each chiller
- c. Evaporator valve for each chiller
- d. Condenser pump for each chiller
- e. Condenser valve for each chiller
- f. Evaporator motorized isolation valve
- g. Condenser motorized isolation valve
- h. Cooling tower fans
- i. Cooling tower valves
- j. Bypass valve

Apart from above equipment following control/measuring field instruments are essential for plant operation and shall be supplied and installed by others.

- 1. Chilled water supply header temperature sensor
- 2. Chilled water return header temperature sensor
- 3. Condenser water supply header temperature sensor
- 4. Condenser water return header temperature sensor
- 5. DP transmitters



4. Operation

Chiller Plant Manager System controls the chiller system and its units which comprises of Evaporator pumps and valves, Condenser pumps and Valves, and Cooling Tower fan and valves.

1. Power on Sequence: -

- MCB's in panel: -
MCB 1 is used for 24VDC SMPS.
MCB 2 & 3 for 24VDC Panel Supply.
MCB 4 is used for Fan, Tube light & 230VAC Aux socket.
- Sequence: -
1st turn ON MCB 1 & 3, then turn ON MCB 2. For Turning OFF follow same procedure in reverse order.

2. Operating Modes: -

Chiller Plant Manager operates in 2 modes, i.e. AUTO/MANUAL. Selection is provided on HMI Panel. For AUTO mode PLC will take care of control process for Chillers. In MANUAL mode individual START/STOP push buttons for individual equipment's are provided on HMI Panel for control.

HMI is used for monitoring & control for process.

HMI screens are as follows: -

- 1) Startup Screen
- 2) Overview
- 3) Evaporator Pumps
- 4) Cooling Tower
- 5) Condenser Pumps
- 6) Chillers
- 7) Settings
- 8) Alarms
- 9) Alarm History
- 10) Modbus Data

Online alarms are provided on top of every screen with reset button on each screen.

2.1. Startup Screen: -

After giving power supply (24VDC) to HMI, HMI will start with the Startup screen as a home screen.



This screen is actual startup screen for the HMI panel. Whenever the Panel is powered ON or the HMI is Powered ON this Screen will be displayed. It has a Startup Time which is activated when PLC is Powered ON. This is a non-settable 60 seconds delay after which PLC starts execution of its program, i.e. it gives its command to equipment's connected to the system.

Startup screen appears on the HMI only for once, if any other screen button is pressed then this screen can be viewed only after power cycle of the HMI. HMI screen Navigation tabs are provided for other screens. Respective screen is shown on pressing the respective tabs.

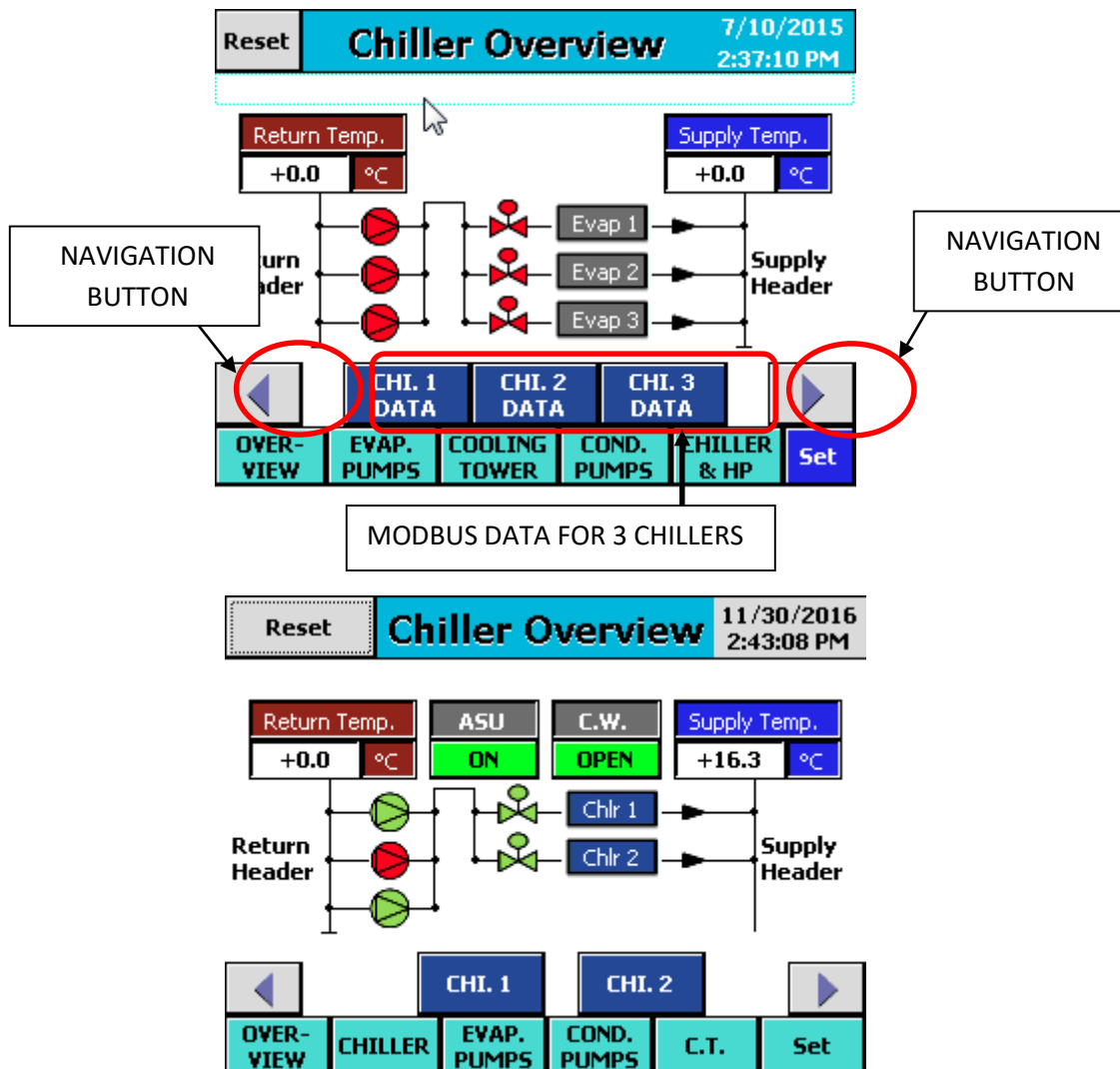
2.2. Overview: -

On pressing the OVERVIEW tab, the following screen will appear on HMI Panel. This screen provides an overview of the system, there are two navigation buttons provided on this screen for navigation of screens within the OVERVIEW screen.

List of screens in OVERVIEW screen: -

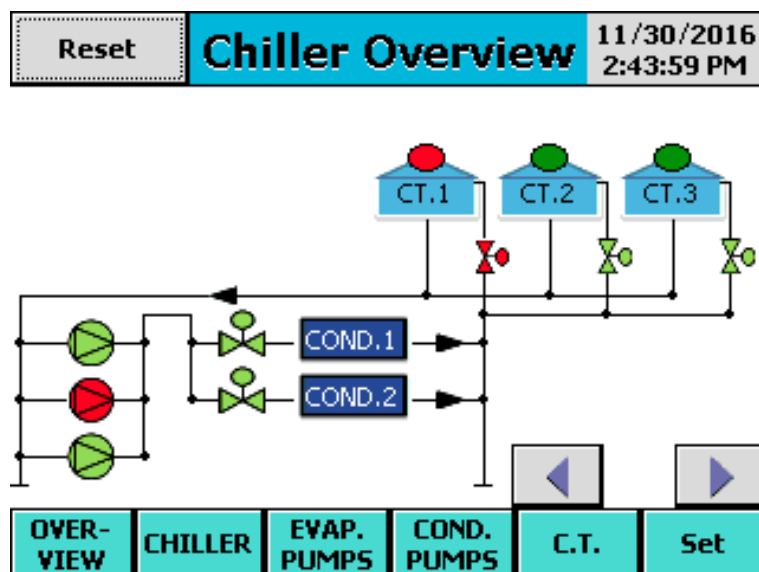
a) Chiller Overview

The following screen shows you status of chiller evaporator equipment, Supply header temp and Return header temp. There 3 buttons provided to view Modbus data of all 3 chillers respectively. Below the screen diagrammatic representation shows the status and their respective color assignments.



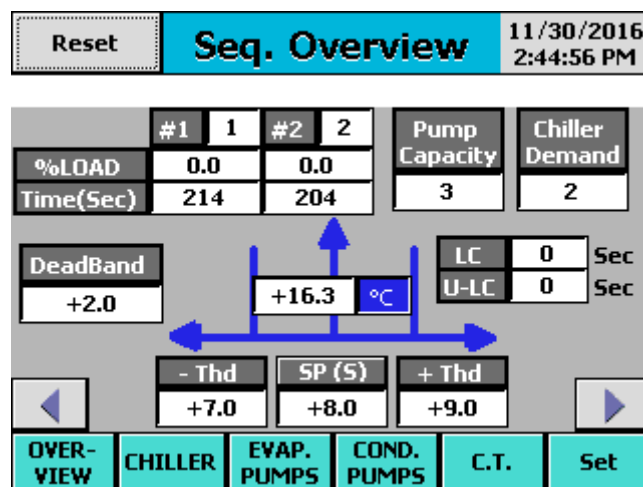
SYMBOL	DESCRIPTION		
	OFF	ON	FAULT
PUMP			
VALVE			
CHILLER			
CT FAN			

The following screen shows you status of chiller condenser equipment and cooling tower equipment.



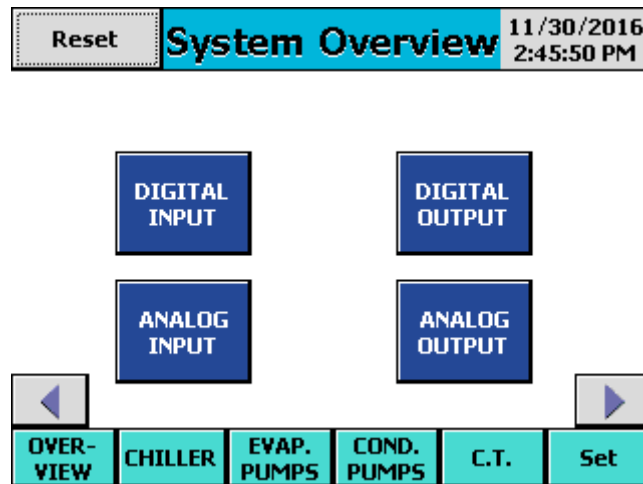
b) Sequence Overview

The following screen shows you the sequence of operation in auto mode. #1, #2 and #3 shows you the sequence of Chillers as per their Run hours and availability. Load and ON Time are also shown for respective chillers. Chiller demand is value of the number of chillers that is required to be ON in present condition. LC is the Load Count Time after which next available chiller will start. U-LC is the Unload Count time after which the next chiller will stop. SP is the setpoint for Supply header temperature. All the fields are for display only; no values can be entered from this screen.



c) System Overview

The following screen shows you the list and current status of all the connected I/O's to the PLC. Each button click will open their respective screen.



i. Digital Input

Reset		Digital Input 1	11/30/2016 4:16:08 PM
I 0.0	CHILLER 1 LOCAL/REMOTE STATUS	CLOSE	
I 0.1	CHILLER 1 RUN STATUS	OPEN	
I 0.2	CHILLER 1 TRIP STATUS	OPEN	
I 0.3	CHILLER 2 LOCAL/REMOTE STATUS	CLOSE	
I 0.4	CHILLER 2 RUN STATUS	OPEN	
I 0.5	CHILLER 2 TRIP STATUS	OPEN	
I 0.6	EVAP VFD ON	OPEN	
I 0.7	COND VFD ON	OPEN	
		BACK	

Reset	Digital Input 2	11/30/2016 4:16:39 PM
I 1.0	CHILLER 3 LOCAL/REMOTE STATUS	CLOSE
I 1.1	CHILLER 3 RUN STATUS	OPEN
I 1.2	CHILLER 3 TRIP STATUS	OPEN
I 1.3	CHLD WP1 LOCAL/REMOTE STATUS	CLOSE
I 1.4	CHILLED WP 1 RUN STATUS	OPEN
I 1.5	CHILLED WP 1 TRIP STATUS	OPEN
I 1.6	CHLR 1 B/F VALVE ON/OFF STATUS	OPEN
I 1.7	SPARE	OPEN
◀ BACK ▶		

Reset	Digital Input 3	11/30/2016 4:17:09 PM
I 2.0	CHLD WP2 LOCAL/REMOTE STATUS	CLOSE
I 2.1	CHILLED WP 2 RUN STATUS	OPEN
I 2.2	CHILLED WP 2 TRIP STATUS	OPEN
I 2.3	CHLR 2 B/F VALVE ON/OFF STATUS	OPEN
I 2.4	CHLD WP3 LOCAL/REMOTE STATUS	CLOSE
I 2.5	CHILLED WP 3 RUN STATUS	OPEN
I 2.6	CHILLED WP 3 TRIP STATUS	OPEN
I 2.7	CHLR 3 B/F VALVE ON/OFF STATUS	OPEN
◀ BACK ▶		

Reset	Digital Input 4	11/30/2016 4:17:42 PM
I 3.0	Chiller PUMP VFD TRIP STATUS	OPEN
I 3.1	COND WP1 LOCAL/REMOTE STATUS	CLOSE
I 3.2	COND. WP 1 RUN STATUS	OPEN
I 3.3	COND. WP 1 TRIP STATUS	OPEN
I 3.4	COND.1 B/F VALVE ON/OFF STATUS	OPEN
I 3.5	ASU Feedback	CLOSE
I 3.6	Chilled Water Valve Feedback	CLOSE
I 3.7	Condenser Pump VFD Trip	OPEN
◀ BACK ▶		

Reset	Digital Input 5	11/30/2016 4:18:15 PM
I 4.0	COND WP2 LOCAL/REMOTE STATUS	CLOSE
I 4.1	COND. WP 2 RUN STATUS	OPEN
I 4.2	COND. WP 2 TRIP STATUS	OPEN
I 4.3	COND.2 B/F VALVE ON/OFF STATUS	OPEN
I 4.4	COND WP3 LOCAL/REMOTE STATUS	CLOSE
I 4.5	COND. WP 3 RUN STATUS	OPEN
I 4.6	COND. WP 3 TRIP STATUS	OPEN
I 4.7	COND.3 B/F VALVE ON/OFF STATUS	OPEN
BACK		

Reset	Digital Input 6	11/30/2016 4:18:41 PM
I 5.0	CT1 FAN RUN STATUS	OPEN
I 5.1	CT1 FAN TRIP STATUS	OPEN
I 5.2	CT1 IN B/F VALVE ON/OFF STATUS	OPEN
I 5.3	CT2 FAN RUN STATUS	OPEN
I 5.4	CT2 FAN TRIP STATUS	OPEN
I 5.5	CT2 IN B/F VALVE ON/OFF STATUS	OPEN
I 5.6	SPARE	OPEN
I 5.7	SPARE	OPEN
BACK		

Reset	Digital Input 7	11/30/2016 4:19:00 PM
I 6.0	CT3 FAN RUN STATUS	OPEN
I 6.1	CT3 FAN TRIP STATUS	OPEN
I 6.2	CT3 IN B/F VALVE ON/OFF STATUS	OPEN
I 6.3	CT_3_PUMP_R/L	OPEN
I 6.4	SPARE	OPEN
I 6.5	SUMMER_SEASON	OPEN
I 6.6	MONSOON_SEASON	OPEN
I 6.7	WINTER_SEASON	OPEN
BACK		

ii. Digital Outputs

Reset	Digital Output 1	11/30/2016 4:19:25 PM
Q 0.0	CHILLER 1 START/STOP Cmd.	OFF
Q 0.1	CHILLER 2 START/STOP Cmd.	OFF
Q 0.2	CHILLER 3 START/STOP Cmd.	OFF
Q 0.3	CHILLED WP 1 STAR/DELTA Cmd.	OFF
Q 0.4	CHILLER 1 B/F VALVE ON/OFF Cmd.	OFF
Q 0.5	CHILLED WP 2 STAR/DELTA Cmd.	OFF
Q 0.6	CHILLER 2 B/F VALVE ON/OFF Cmd.	OFF
Q 0.7	SPARE	OFF
◀ BACK ▶		

Reset	Digital Output 2	11/30/2016 4:19:47 PM
Q 1.0	CHILLED WP 3 STAR/DELTA Cmd.	OFF
Q 1.1	CHILLER 3 B/F VALVE ON/OFF Cmd.	OFF
Q 1.2	VFD ON/OFF Cmd.	OFF
Q 1.3	COND.WP 1 STAR/DELTA Cmd.	OFF
Q 1.4	COND. 1 B/F VALVE ON Cmd.	OFF
Q 1.5	COND.WP 2 STAR/DELTA Cmd.	OFF
Q 1.6	COND. 2 B/F VALVE ON Cmd.	OFF
Q 1.7	COND.WP 3 VFD Cmd.	OFF
◀ BACK ▶		

Reset	Digital Output 3	11/30/2016 4:20:10 PM
Q 2.0	COND.WP 3 STAR/DELTA Cmd.	OFF
Q 2.1	COND. 3 B/F VALVE ON Cmd.	OFF
Q 2.2	CT1 FAN START/STOP Cmd.	OFF
Q 2.3	CT1 B/F VALVE ON/OFF Cmd.	OFF
Q 2.4	CT2 FAN START/STOP Cmd.	OFF
Q 2.5	CT2 B/F VALVE ON/OFF Cmd.	OFF
Q 2.6	CT3 FAN START/STOP Cmd.	OFF
Q 2.7	CT3 B/F VALVE ON/OFF Cmd.	OFF
◀ BACK ▶		

Reset	Digital Output 4	11/30/2016 4:20:36 PM
Q 3.0	Chiller 1 Run Status	OFF
Q 3.1	Chiller 2 Run Status	OFF
Q 3.2	Chiller Trip Status	OFF
Q 3.3	CHILLED WP 1 VFD Cmd.	OFF
Q 3.4	CHILLED WP 2 VFD Cmd.	OFF
Q 3.5	CHILLED WP 3 VFD Cmd.	OFF
Q 3.6	COND.WP 1 VFD Cmd.	OFF
Q 3.7	COND.WP 2 VFD Cmd.	OFF
◀ BACK ▶		

iii. Analog Input

Reset

Analog Input 1

11/30/2016
4:20:59 PM

PIW 8	CHLR SUP. HEADER TEMP.	+16.3	Deg C
PIW 10	CHLR RET. HEADER TEMP	+0.0	Deg C
PIW 12	CHILLER FLOW AT SUPPLY	+0.0	GPM
PIW 14	VFD FEEDBACK	+0.0	RPM

◀

BACK

▶

Reset

Analog Input 2

11/30/2016
4:21:18 PM

PIW 16	AHU 1 PRESSURE	+0.0	psig
PIW 18	AHU 2 PRESSURE	+0.0	psig
PIW 20	SPARE	+0.0	---
PIW 22	SPARE	+0.0	---

◀

BACK

▶

iv. Analog Output

Reset		Analog Output		11/30/2016 4:22:46 PM
PQW 8	VFD_COMMAND	40	%	
PQW 10	CONDENSER_FAN_SPEED	0	---	
PQW 12	SPARE	0	---	
PQW 14	SPARE	0	---	

◀
BACK
▶

2.3. Evaporator Pumps: -

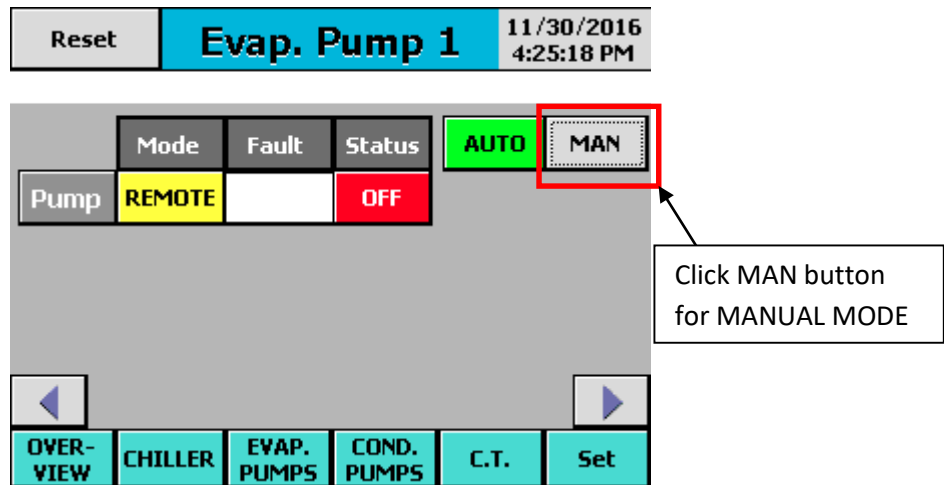
Evaporator screen shows the conditions of the evaporator pump related to Pump 1, Pump 2 and Pump3. It indicates the fault which is generated during the operation. When Mode is LOCAL no operation of the respective pump is possible in Auto or Manual mode. Navigation buttons are to be used to navigate through the screen for Pump 2 and Pump 3.

i. Evaporator Pump 1

Reset		Evap. Pump 1		11/30/2016 4:24:39 PM
	Mode	Fault	Status	AUTO MAN
Pump	LOCAL		OFF	

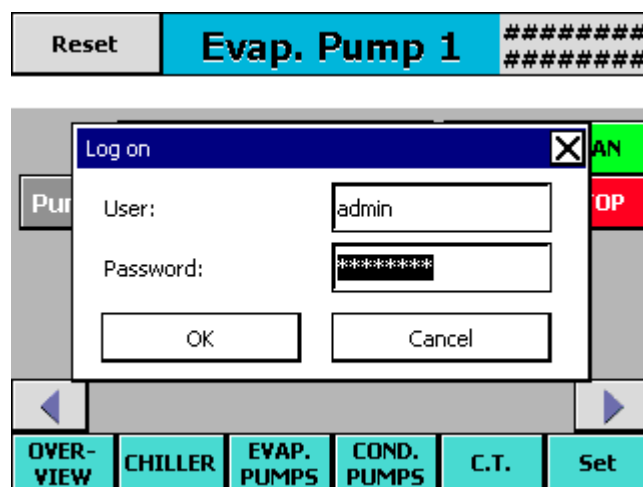
◀
▶

OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set
-----------	---------	-------------	-------------	------	-----



To change the operation of the pump from AUTO to MANUAL click on MANUAL button, a LOG ON window will appear on the screen. If you have Authorization for this level of access, then enter your ID and Password.

It is recommended to be very careful while switching from Auto to manual when device is ON.



If the device is ON, then after switching from AUTO to MANUAL it will remain ON due to BUMPLESS TRANSFER. No two device of same type should be transferred from AUTO to MANUAL at a time.

Reset	Evap. Pump 1			11/30/2016 4:26:05 PM	
	Mode	Fault	Status	AUTO	MAN
Pump	REMOTE		ON	START	STOP
◀ ▶					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

Condition 1 - When 1 or 2 chillers are ON: -

Switching the ON device from Auto to Manual will turn the next OFF device in sequence ON, and the chiller demand will remain same.

Condition 2 - When 3 chillers are ON: -

Switching the ON device from Auto to Manual, a settable delay timer will start counting before which all the other device of one respective sequence should be transferred from Auto to Manual by the operator, failing to do so will cause the tripping of chiller. Chiller demand is also decreased.

ii. Evaporator Pump 3

Reset	Evap. Pump 3			11/30/2016 2:50:36 PM	
	Mode	Fault	Status	AUTO	MAN
Pump	REMOTE		ON		
◀ ▶					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

If there is a fault while the device is running, then the device will go in fault as shown below and shows the cause by which it is in fault.

Reset		Evap. Pump 1		11/30/2016 4:57:32 PM	
11/30/2016 4:57:19 PM Evap1_Pump_Fail to Run					
	Mode	Fault	Status	AUTO	MAN
Pump	REMOTE	FTR	FAULT	START	STOP
◀		▶			
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

2.4. Cooling Tower: -

Cooling Tower screen shows the status of CT fan, CT Inlet valve and CT Outlet Valve of respective sequence.

i. Cooling Tower 1

Reset		Cooling Tower 1		12/1/2016 11:49:55 AM	
	Mode	Fault	Status	AUTO	MAN
Valve			CLOSE		
Fan			OFF		
◀ ▶					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

ii. Cooling Tower 2

Reset		Cooling Tower 2		12/1/2016 11:50:25 AM	
	Mode	Fault	Status	AUTO	MAN
Valve			OPEN		
Fan			ON		
◀ ▶					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

iii. Cooling Tower 3

Reset	Cooling Tower 3			12/1/2016 11:50:38 AM	
	Mode	Fault	Status	AUTO	MAN
Valve			OPEN		
Fan			ON		
◀ ▶					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

Reset	Cooling Tower 3			12/1/2016 11:51:56 AM	
12/1/2016 11:51:52 AM CT3_Fan_Trip					
	Mode	Fault	Status	AUTO	MAN
Valve			CLOSE		
Fan		TRIP	FAULT		
◀ ▶					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

2.5. Condenser Pump: -

Condenser screen shows the status of Condenser pumps and Heat pump condenser on respective screens.

i. Condenser Pump 1

Reset	Cond. Pump 1			12/1/2016 11:52:36 AM	
	Mode	Fault	Status	AUTO	MAN
Pump	REMOTE		ON		
◀ ▶					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

ii. Condenser Pump 2

Reset	Cond. Pump 2			12/1/2016 11:52:53 AM	
	Mode	Fault	Status	AUTO	MAN
Pump	REMOTE		OFF		
◀ ▶					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

iii. Condenser Pump 3

Reset	Cond. Pump 3			12/1/2016 11:53:06 AM	
	Mode	Fault	Status	AUTO	MAN
Pump	REMOTE		ON		
◀ ▶					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

2.6. Chillers: -

Chiller screen shows the status of Chillers, Evaporator valve, Condenser valve, and Heat pump evaporator on respective screens.

i. Chiller 1

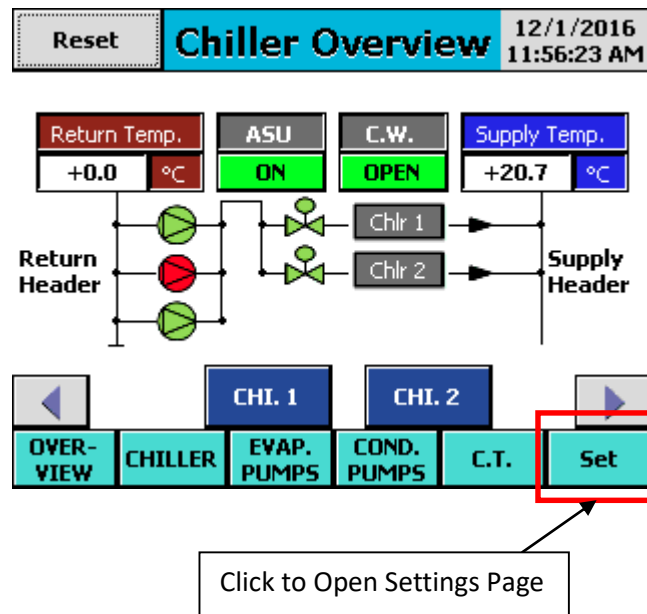
Reset		Chiller 1		12/1/2016 11:53:56 AM	
	Mode	Fault	Status	AUTO	MAN
Eva-V			OPEN		
Cnd-V			OPEN		
Chiller	REMOTE		OFF		
◀ ▶					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

ii. Chiller 2

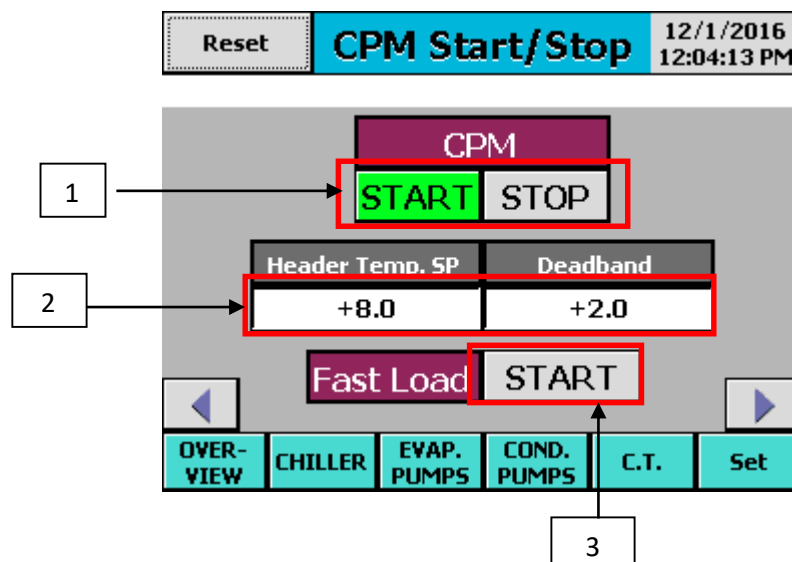
Reset		Chiller 2		12/1/2016 11:54:10 AM	
	Mode	Fault	Status	AUTO	MAN
Eva-V			OPEN		
Cnd-V			OPEN		
Chiller	REMOTE		OFF		
◀ ▶					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

2.7. Settings: -

Settings screen shows the setting of various parameters for operation of the plant manager in AUTO Mode. Click on Set button to open Settings Page.

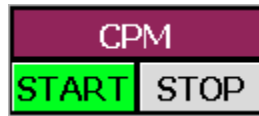


i. CPM Start/Stop



1. Plant Manager Start/Stop Button

Operation of Plant Manager in AUTO Mode is only possible when the START button is pressed; else AUTO operation for plant manager will not be executed. When Plant manager is in Auto and demand is 0 then Evaporator pump 1 and evaporator valve1 are ON for circulation of air.



2. Header Temperature SP and Dead band

When Header temperature is greater than Header Temperature SP + Dead band/2 then the plant manager is in LOADING ZONE and when Header temperature is less than Header Temperature SP- Dead band/2 then the plant manager is in UNLOADING ZONE.

LOADING ZONE – When plant manager is in loading zone then chiller units are turned ON as per the time settings, sequence and availability of the chillers.

Start Sequence / Loading Sequence: -

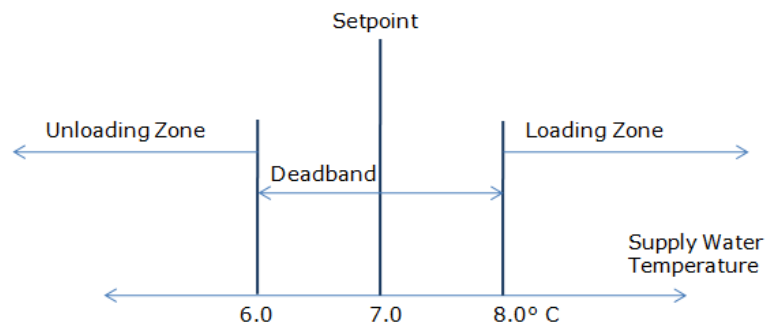
- **Start**
- **Evaporator Pump**
 - Preset Delay
- **Evaporator Valve**
 - Preset Delay
- **Condenser Pump**
 - Preset Delay
- **Condenser Valve**
 - Preset Delay
- **CT valve**
 - Preset Delay
- **Chiller**
 - Preset Delay
- **CT Fan**

UNLOADING ZONE – When plant manager is in unloading zone then chiller units are turned OFF as per the time settings and sequence of the chillers.

Stop Sequence / Unloading Sequence

- **Stop**
- **Chiller**

- Preset Delay
- **CT Valve**
 - Preset Delay
- **Condenser Valve**
 - Preset Delay
- **CT Fan**
 - Preset Delay
- **Condenser Pump**
 - Preset Delay
- **Evaporator Valve**
 - Preset Delay
- **Evaporator Pump**



3. Fast Load Start Button

Fast Load start button is used when the plant manager is in loading zone and the timer for next chiller ON is counting, then on pressing this button the counting will be bypassed and the start sequence for next chiller in sequence will be started.

- ii. Set Point
 - a) Set Point – 1

Reset	Set Point - 1		12/1/2016 12:16:33 PM
% Load Minimum	65.0	%	1
% Load Max	95.0	%	2
Next Chi. ON Delay time	15	min	3
ON Delay SP	5	sec	4
OFF Delay SP	5	sec	5

OVER-VIEW CHILLER EVAP. PUMPS COND. PUMPS C.T. Set

1. % Load Min – 2 Chi ON

When 2 chillers are ON and the plant manager goes in Unloading zone then when the % load of the ON chillers goes below this SP then the stop sequence for next Chiller in sequence will start after the count of OFF delay SP is completed.

2. % Load Max

When Plant Manager is in Loading zone and % Load of the ON chiller goes above this SP then the sequence to start the next chiller in sequence is started after the count of ON Delay SP.

3. Next Chi. ON Delay time

If the Plant Manager is in loading zone and ON Chiller % loading does not reach the % Load Max SP for this Specified time, then the sequence to start the next chiller in sequence is started after the count of ON Delay SP. Set Point is in minutes.

4. ON Delay SP

When the ON Chiller % loading is more then % Load SP or Next Chi. ON Delay time is elapsed then the plant manager counts the set minutes and after the time is elapsed, Plant manager starts the ON sequence for next chiller. Set Point is in minutes.

5. OFF Delay SP

When Plant manager goes in Unloading zone then it counts the set minutes and after the time is elapsed, Plant manager starts the stop sequence for next chiller in sequence. Set Point is in minutes.

b) Set Point 2

Reset		Set Point - 2		12/1/2016 12:30:17 PM	
AUTO SP	1.0	bar	P	0.1	---
MANUAL SP	1.0	%	I	150	---
VFD SPEED Max.	100.0	%	D	5	---
VFD SPEED Min.	40.0	%			
PID	AUTO		MANUAL		
OVER-VIEW CHILLER EVAP. PUMPS COND. PUMPS C.T. Set					

c) Set Point – 3

Reset		Set Point - 3		12/1/2016 12:43:45 PM	
Cond. IN Temp SP	+0.0	Deg C			
ON Hyst	0	Deg C			
Man off delay	120	Sec			
Chiller Capacity	60.0	%			
OVER-VIEW CHILLER EVAP. PUMPS COND. PUMPS C.T. Set					

1. Condenser IN Temp SP and ON Hyst

When the Condenser water in temperature is below the Temp SP then the CT fan will not be started and When the Condenser water in temperature is above Temp SP +ON Hyst then only CT fan will start. Set Point is in degree Celsius.

2. Manual OFF Delay

When all three chillers are running and any one of device which is running in auto mode, is switched to Manual mode then this delay comes in picture. The timers start counting for the set time, before the timer is elapsed all the device of any one sequence should be switched to Manual mode, else it will cause tripping of the chiller. Set Point is in seconds.

3. Chiller Capacity

The following screen shows you the Analog Input scaling of the analog inputs that are connected to the system. High and Low range of the device can be given here as per the sensors connected in field. Offset value for each Analog Input is given, you can enter the value of offset for respective Analog Input in here and that value will be added to the actual value.

d) Setpoints-4

Reset	Set Point - 4		12/1/2016 12:50:34 PM
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Summer SP	+7.0	Deg C
Monsoon SP	+7.0	Deg C
Winter SP	+7.0	Deg C

◀
▶

OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set
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Based on Input provided Seasonal Set points are applied to CPM.

iii. Analog Scaling

The following screen shows you the Analog Input scaling of the analog inputs that are connected to the system. High and Low range of the device can be given here as per the sensors connected in field. Offset value for each Analog Input is given, you can enter the value of offset for respective Analog Input in here and that value will be added to the actual value.

Reset		Analog Input		12/1/2016 12:52:42 PM	
		CHILLER TEMP SUPPLY RETURN		CHILLER FLOW	VFD FEEDBACK
Range HIGH	+100.0	+100.0	+100.0	+100.0	+100.0
Range LOW	+0.0	+0.0	+0.0	+0.0	+0.0
OFFSET	+0.0	+0.0	+0.0	+0.0	+0.0
◀ ▶					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

Reset		Analog Input		12/1/2016 12:54:18 PM	
		AHU PRESSURE 1 2		SPARE	SPARE
Range HIGH	+100.0	+100.0	+100.0	+100.0	+100.0
Range LOW	+0.0	+0.0	+0.0	+0.0	+0.0
OFFSET	+0.0	+0.0	+0.0	+0.0	+0.0
◀ ▶					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

iv. Chiller Valve Setting

The following screen shows you the Valve feedback time setting for respective Evaporator valve, Condenser valve, CT inlet valve and CT outlet valve. When the output command is close and the feedback for valve close is not received by the system before the time is elapsed then the respective valve will go in fault i.e. FTC (Fail to Close). And when the output command is open and the feedback for valve open is not received by the system before the time is elapsed then the respective valve will go in fault i.e. FTO (Fail to Open). Set Point is in seconds.

Reset

Chi. Valve Set.

12/1/2016

12:54:37 PM

Time In Seconds	Eva.	Cond.	CT
Chi.1 Valve f/b Timer	10	10	10
Chi.2 Valve f/b Timer	10	10	10

◀

▶

OVER-VIEW

CHILLER

EVAP. PUMPS

COND. PUMPS

C.T.

Set

v. Chiller Pump Setting

The following screen shows you the Pump ON feedback time and OFF feedback time setting for respective Chiller, Evaporator Pump, Condenser Pump, CT Fan and HP Pumps. The timing for Pump ON and OFF is separate and respective for each Pump. When the output command is stop and the feedback for Pump stop is not received by the system before the OFF time is elapsed then the respective Pump will go in fault i.e. FTS (Fail to Stop). And when the output command is start and the feedback for Pump run is not received by the system before the ON time is elapsed then the respective Pump will go in fault i.e. FTR (Fail to Run). Set Point is in seconds.

Reset

Chi1 & Pump Set.

12/1/2016
12:54:53 PM

Time In Seconds	Chiller	Evap_P	Cond_P	CT_Fan
ON f/b Timer	70	10	10	10
OFF f/b Timer	70	10	10	10

◀

▶

OVER-VIEWCHILLEREVAP. PUMPSCOND. PUMPSC.T.Set

Reset

Chi2 & Pump Set.

12/1/2016
12:55:20 PM

Time In Seconds	Chiller	Evap_P	Cond_P	CT_Fan
ON f/b Timer	70	10	10	10
OFF f/b Timer	70	10	10	10

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OVER-VIEWCHILLEREVAP. PUMPSCOND. PUMPSC.T.Set

Reset

Pump Set. 3

12/1/2016
12:57:46 PM

Time In Seconds	Evap_P	Cond_P
ON f/b Timer	10	10
OFF f/b Timer	10	10

◀

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OVER-VIEWCHILLEREVAP. PUMPSCOND. PUMPSC.T.Set

vi. Delay Setting

The following screen shows you Delay time for start sequence and for stop sequence the delay for respective valve and pump to start and stop are displayed in the screen which can be changed as per requirement. If operator requires that there should be no delay between the two devices to start or stop then the ON or OFF delay for second device should be made to zero. Set Point is in seconds.

Reset
Valve Delay Set.
12/1/2016
12:58:00 PM

Time In Seconds	Evap_V	Cond_V	CT_V
ON Delay Time	10	10	10
OFF Delay Time	10	10	10

◀
▶

OVER-VIEW
CHILLER
EVAP. PUMPS
COND. PUMPS
C.T.
Set

Reset
Pump Delay Set.
12/1/2016
12:58:23 PM

Time In Seconds	Chiller	Evap_P	Cond_P	CT_Fan
ON Delay Time	10	10	10	10
OFF Delay Time	10	10	10	10

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OVER-VIEW
CHILLER
EVAP. PUMPS
COND. PUMPS
C.T.
Set

vii. Set Date and Time

The following screen allows you set the date and time of the HMI if it lags or leads the actual time. Enter the value in the input field as shown in the block and then press Set button to overwrite the current date and time of the HMI.

Reset **Set Date & Time** 12/1/2016 12:59:33 PM

mm/dd/yyyy hh:mm:ss
12/1/2016 1:00:00 PM

Set

Press Set button to overwrite current date and time

← →

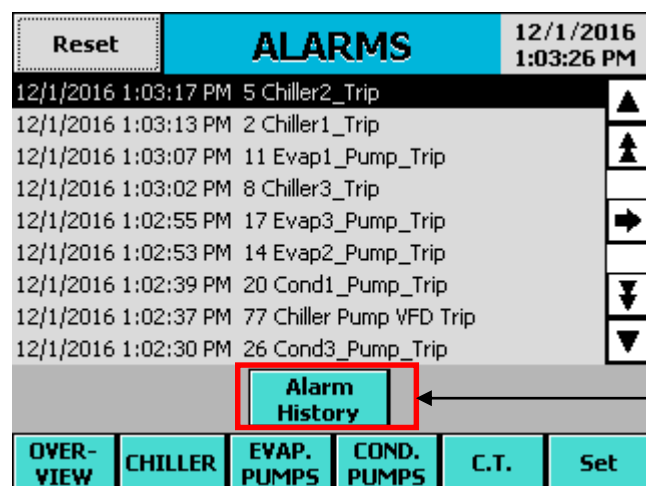
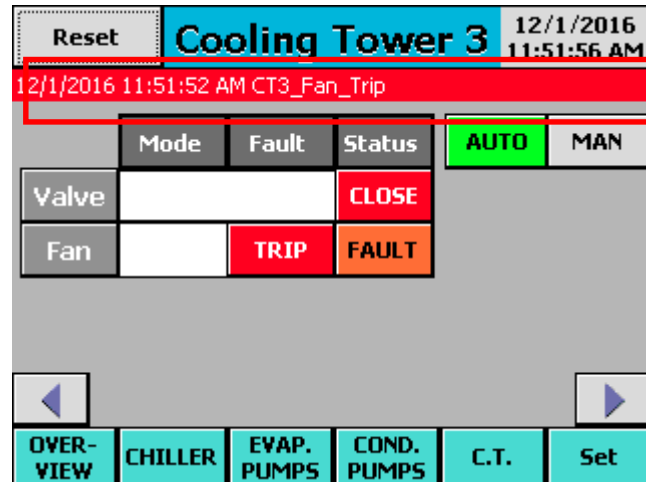
OVER-VIEW CHILLER EVAP. PUMPS COND. PUMPS C.T. Set

7/17/2015

A	B	C	D	E	F	G	H	I	J
K	L	M	N	O	P	Q	R	S	T
U	V	W	X	Y	Z	/	*	-	+
.	:	,	=	_	(	)	@	"	
0	1	2	3	4	5	6	7	8	9
Shift	←	→	BSP			ESC	↵		

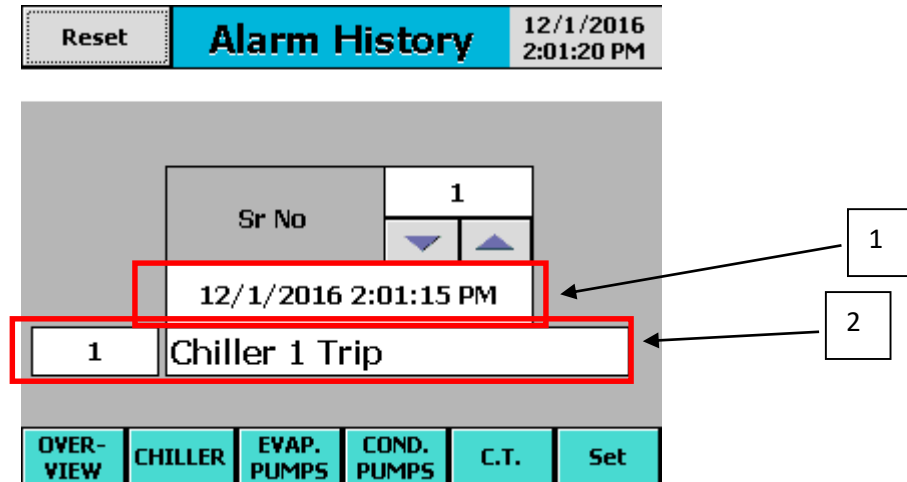
2.8. Alarm: -

Alarm screen shows the current alarm of the system.



2.9. Alarm History: -

The following Screen shows Previous Alarms.



1. Date & Time :-

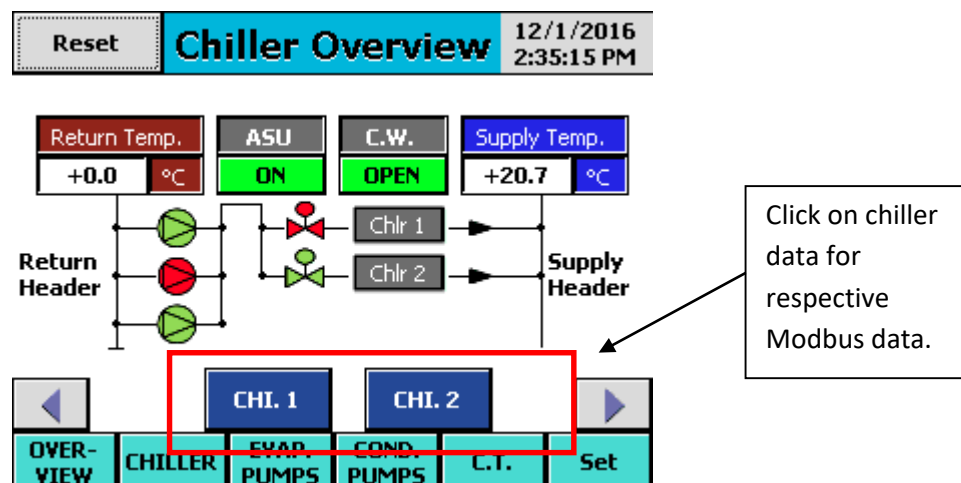
This field Shows Alarm Date & Time.

2. Message :-

This field shows alarm text along with alarm number.

2.10. Modbus Data:-

Modbus data of the chillers will be displayed in the following screen.



Reset	Chiller 1 Data	12/1/2016 2:35:39 PM
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1	Evap. Water In Temp.	+0.0	°C
2	Evap. Water Out Temp.	+0.0	°C
3	Sump Oil Temp.	+0.0	°C
4	Discharge Temp.	+0.0	°C
5	Suction Temp.	+0.0	°C

▲	Page 1 of 4				▼
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

Reset	Chiller 1 Data	12/1/2016 2:36:07 PM
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6	Liquid Line Temp.	+0.0	°C
7	Cond. Water In Temp.	+0.0	°C
8	Cond. Water Out Temp.	+0.0	°C
9	Suction Pressure	+0.0	bar
10	Discharge Pressure	+0.0	bar

▲	Page 2 of 4				▼
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

Reset	Chiller 1 Data	12/1/2016 2:36:22 PM
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11	Oil Feed Pressure	+0.0	bar
12	Motor Current	+0.0	A
13	Suction Superheat	+0.0	°C
14	Comp. Current In %	+0.0	%
15	Chiller RunHour	3	HH

▲	Page 3 of 4				▼
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

Reset	Chiller 1 Data	12/1/2016 2:36:33 PM
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16	Ready to Start	0	
17		0	
18		0	
19		0	
20		0	

	Page 4 of 4					
OVERVIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set	

Reset	Chiller 2 Data	12/1/2016 2:36:50 PM
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1	Evap. Water In Temp.	+0.0	°C
2	Evap. Water Out Temp.	+0.0	°C
3	Sump Oil Temp.	+0.0	°C
4	Discharge Temp.	+0.0	°C
5	Suction Temp.	+0.0	°C

	Page 1 of 4					
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set	

Reset	Chiller 2 Data	12/1/2016 2:37:04 PM
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6	Liquid Line Temp.	+0.0	°C
7	Cond. Water In Temp.	+0.0	°C
8	Cond. Water Out Temp.	+0.0	°C
9	Suction Pressure	+0.0	bar
10	Discharge Pressure	+0.0	bar

	Page 2 of 4					
OVERVIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set	

Reset	Chiller 2 Data	12/1/2016 2:37:14 PM
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11	Oil Feed Pressure	+0.0	bar
12	Motor Current	0.0	A
13	Suction Superheat	+0.0	°C
14	Comp. Current In %	+0.0	%
15	Chiller RunHour	0	HH

▲	Page 3 of 4				▼
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set

Reset	Chiller 2 Data	12/1/2016 2:37:36 PM
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16	Ready to start	0	
17		0	
18		0	
19		0	
20		0	

▲	Page 4 of 4				▼
OVER-VIEW	CHILLER	EVAP. PUMPS	COND. PUMPS	C.T.	Set